PUSHOUTS ALONG CLOSED IMMERSIONS OF FINITE TYPE PRESERVE FINITE TYPE OVER A LOCALLY NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19861-19868 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19870-19873, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The cited Situation and Proposition labels exist in the pinned source and resolve at assembly; neither is carded by any lane yet.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter situation pushout-along-closed-immersion-and-integral and proposition of the same name (both in-source, uncarded); More on Algebra, lemma more-algebra-lemma-fibre-product-finite-type.

Source reference. Original lemma 00IT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19873–19873. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E25: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19758–19761. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19716. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0ECI assume S is a locally Noetherian scheme and X, Y, and Z are locally of finite type over S. Then the pushout $Y \amalg_Z X$ (Proposition 0E25) is locally of finite type over S.*

Proof. Looking on affine opens we recover the result of More on Algebra, Lemma 00IT. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.